

SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT NAME : AI

- A) Implement the Breadth First Search algorithm to solve a given problem.
- B) Implement the Iterative Depth First Search algorithm to solve the same problem.
- C) Compare the performance and efficiency of both algorithms.

CODE:

```
print("T090_MADESH SIVAKUMAR")
```

```
from collections import deque
```

```
# Graph represented as an adjacency list
```

```
city_map = {
```

```
    'College': ['Bus Stop', 'Library'],
```

```
    'Bus Stop': ['Market'],
```

```
    'Library': ['Hospital'],
```

```
    'Market': ['Railway Station'],
```

```
    'Hospital': ['Mall'],
```

```
    'Railway Station': ['Airport'],
```

```
    'Mall': ['Airport'],
```

```
    'Airport': []
```

```
}
```

```
# A) Breadth First Search (BFS)
```

```
def bfs_shortest_path(graph, start, destination):
```

```
    queue = deque([(start, [start])])
```

```
    visited = set([start])
```

```
    nodes_explored_count = 0
```

```
    while queue:
```

```
        current, path = queue.popleft()
```

```
        nodes_explored_count += 1
```

if current == destination:

return path, nodes_explored_count

for neighbor in graph.get(current, []):

if neighbor not in visited:

visited.add(neighbor)

queue.append((neighbor, path + [neighbor]))

return None, nodes_explored_count

B) Iterative Depth First Search (DFS)

def iterative_dfs_path(graph, start, destination):

stack = [(start, [start])]

visited = set()

nodes_explored_count = 0

while stack:

current, path = stack.pop()

nodes_explored_count += 1

if current == destination:

return path, nodes_explored_count

if current not in visited:

visited.add(current)

for neighbor in graph.get(current, []):

if neighbor not in visited:

stack.append((neighbor, path + [neighbor]))

return None, nodes_explored_count

Execution

start_node = 'College'

end_node = 'Airport'

bfs_path, bfs_count = bfs_shortest_path(city_map, start_node, end_node)

dfs_path, dfs_count = iterative_dfs_path(city_map, start_node, end_node)

print("----- BFS Results -----")

print("Path Found:", " -> ".join(bfs_path))

print("Total Steps (Edges):", len(bfs_path) - 1)

print("Total Nodes Visited:", bfs_count)

print("\n----- DFS Results -----")

print("Path Found:", " -> ".join(dfs_path))

print("Total Steps (Edges):", len(dfs_path) - 1)

print("Total Nodes Visited:", dfs_count)

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OUTPUT:

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/T090_MADESH/Artificial Intelligence/T090_AI_PRACTICAL1.py =====
T090_MADESH SIVAKUMAR
----- BFS Results -----
Path Found: College -> Bus Stop -> Market -> Railway Station -> Airport
Total Steps (Edges): 4
Total Nodes Visited: 8

----- DFS Results -----
Path Found: College -> Library -> Hospital -> Mall -> Airport
Total Steps (Edges): 4
Total Nodes Visited: 5
>>> |
```